LOCALLY CONSTANT SHEAVES OF FINITE TYPE ON THE AFFINE LINE OVER A NOETHERIAN AFFINE BASE HAVE CONSTRUCTIBLE LOWER SHRIEK

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Allowed standard prerequisites (editorial).

- Finite etale covering trivializing locally constant sheaves (Etale Cohomology, Lemma etale-cohomology-lemma-pullback-filtered-modules, canonical tag 0GJ0). cited at proof lines 4848-4850. Supplies the finite etale covering g : U' -> U over which g^{-1}F is filtered by constant sheaves.
- Trace method section (Etale Cohomology, Section etale-cohomology-section-trace-method, canonical tag 03SH). cited at proof lines 4851-4853. Provides the maps F -> g_*g^{-1}F -> F whose composition is an isomorphism.
- Composition of derived lower shriek (Lemma lemma-shriek-composition, canonical tag 0F7J). cited at proof lines 4859-4860. Gives Rf_!g_*g^{-1}F = R(f after g)_!g^{-1}F.
- Constant sheaves case (Lemma lemma-constant-shriek-rel-dim-1, canonical tag 0GKX). cited at proof lines 4864-4865. The target case reached by the finite filtration of g^{-1}F by constant sheaves.

Source reference. Original lemma 0GJ0: sources/stacks-chow/tags.txt, source lines 20635–20635. Cited at proof lines 4848-4850: supplies the covering g : U' -> U over which g^{-1}F has a finite filtration by constant sheaves.

Source reference. Original section 03SH: sources/stacks-chow/tags.txt, source lines 4968–4968. Cited at proof lines 4851-4853: its discussion gives the maps F -> g_*g^{-1}F -> F whose composition is an isomorphism.

Source reference. Original lemma 0F7J: sources/stacks-chow/tags.txt, source lines 19009–19009. Cited at proof lines 4859-4860: gives Rf_!g_*g^{-1}F = R(f after g)_!g^{-1}F.

Source reference. Original lemma 0GKX: sources/stacks-chow/tags.txt, source lines 20702–20702. Cited at proof lines 4864-4865: the constant sheaves case reached by the finite filtration of g^{-1}F.

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Lemma 0.2. *Let Y be a Noetherian affine scheme of finite dimension. Let Λ be a Noetherian ring which is torsion. Let $\mathcal{F}$ be a finite type, locally constant sheaf of Λ-modules on an open subscheme $U \subset \mathbf{A}^1_Y$. Then $Rf_!\mathcal{F}$ has constructible cohomology sheaves where $f : U \to Y$ is the structure morphism.*

Proof. We may decompose Λ as a product $\Lambda = \Lambda_1 \times \ldots \times \Lambda_r$ where Λ_i is ℓ_i-primary for some prime ℓ_i. Thus we may assume there exists a prime ℓ and an integer $n > 0$ such that ℓ^n annihilates Λ (and hence $\mathcal{F}$).

Since U is Noetherian, we see that U has finitely many connected components. Thus we may assume U is connected. Let $g : U' \to U$ be the finite étale covering constructed in Étale Cohomology, Lemma 0GJ0. The discussion in Étale Cohomology, Section 03SH gives maps

$$\mathcal{F} \to g_* g^{-1}\mathcal{F} \to \mathcal{F}$$

whose composition is an isomorphism. Hence it suffices to prove the result for $g_* g^{-1}\mathcal{F}$. On the other hand, we have $Rf_! g_* g^{-1}\mathcal{F} = R(f \circ g)_! g^{-1}\mathcal{F}$ by Lemma 0F7J. Since $g^{-1}\mathcal{F}$ has a finite filtration by constant sheaves of Λ-modules of the form $\underline{M}$ for some finite Λ-module M (by our choice of g) this reduces us to the case proved in Lemma 0GKX. □